

Exercise 1: SQL fundamentals

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Course: Data Analytics

Sql Fundamentals: SELECT And Filtering.

01. Selecting And Sorting

01. Retrieve all columns

SELECT *

FROM employees;

Output

id	first_name	last_name	department	Salary	hire_date	city
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Makoenq	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

02. Unique departments

SELECT DISTINCT department

FROM employees;

Outcome

department
IT
HR
Finance
Marketing

Q3. Names ordered by salary (DESC)

```
SELECT first_name,  
       last_name  
FROM employees  
ORDER BY salary DESC;
```

Outcome

first_name	last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Molcaena
Brian	Lee

Q4. Top 3 highest-paid employees

```
SELECT id, first_name, last_name, salary  
FROM employees  
ORDER BY salary  
LIMIT 3;
```

Output

id	first_name	last_name
5	Eva	Naidoo
1	Alice	Green
3	Cathy	Zulu

02. Filtering with WHERE

05. Employees in IT

```
SELECT id, first-name, last-name  
FROM employees  
WHERE department = 'IT';
```

Output

id	first-name	last-name
1	Alice	Green
5	Eva	Naidoo

06. Finance with salary > 6000

```
SELECT id, first-name, last-name, department  
FROM employees  
WHERE department = 'Finance'  
AND salary > 6000;
```

Output

id	first-name	last-name	department	salary
3	Cathy	Zulu	Finance	65000

07. Employees in HR or Marketing

```
SELECT id, first-name, last-name, department  
FROM employees  
WHERE department = 'HR'  
OR department = 'Marketing';
```

id	first_name	last_name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

08. Employees not in IT

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department != 'IT';
```

Outcome

id	first_name	last_name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

09. Employees in IT, HR or Finance

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department IN ('IT', 'HR', 'Finance');
```

Outcome

id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

10. Employees in IT with salary > 65000 and city Johannesburg

```
SELECT id, first_name, last_name, department,
       salary, city
FROM employees
WHERE department = 'IT'
AND salary > 65000
AND city = 'Johannesburg';
```

Outcome

id	first_name	last_name	department	Salary
1	Alice	Green	IT	70000

city
Johannesburg